

Pratik Chavan

AI/ML Engineer || Conversational AI Specialist

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🌐 Portfolio

Profile Summary

Software Engineer specializing in Conversational AI and intelligent system implementation, with production experience delivering **LLM-powered chatbots, agentic AI workflows, and RAG-based solutions**. Skilled in prompt engineering, API integration, structured output design, and end-to-end deployment of AI-driven applications. Experienced in direct client-facing technical roles — gathering requirements, validating solutions on-site, and resolving production issues. Proficient in **LangChain, LangGraph, FastAPI**, and multi-agent orchestration patterns including tool calling, MCP-style interfaces, and function-calling agents.

Work Experience

AI/ML Engineer

Jan 2025 – Present

Simulytics Services Pvt Ltd | Full-time

Pune

- **[Conversational AI]** Designed and integrated an AI-powered chatbot into a commercial SaaS Warehouse Management Platform, enabling end-users to interact with warehouse data, order queries, and operational workflows through natural language.
- Implemented conversational flows using **LangChain** with conversation memory, prompt templates, and structured output parsers — ensuring reliable, typed responses for downstream business process consumption.
- Applied prompt engineering best practices: few-shot prompting, chain-of-thought, output formatting constraints, and fallback handling to **reduce hallucination rates** in production.
- Built and deployed **RESTful AI inference APIs using FastAPI**; integrated authentication (JWT), rate limiting, and error-handling patterns to meet production-grade SaaS quality standards.
- Containerized AI services with **Docker** for consistent staging-to-production deployments; managed end-to-end service delivery including testing, validation, and go-live support.
- Led **on-site client engagements** across multiple customer environments: gathered technical requirements, conducted solution demonstrations, diagnosed integration issues, and ensured smooth go-live.
- Developed and maintained integrations using REST APIs and modern web technologies (JavaScript, React); collaborated cross-functionally with product, engineering, and domain experts.
- Implemented **computer vision features** (carton counting, PPE detection, barcode scanning) as microservices; optimized models using ONNX/TFLite for on-device Android inference via Retrofit API.
- Actively involved in SaaS WMS product planning and frontend implementation using React; gained working knowledge of commercial SaaS product architecture, billing structures, and iterative feature delivery.

Web Development Intern

Jan 2025 – Jun 2025

Skill Encoder

- Developed a React Native lead management application with MongoDB backend; worked in agile cross-functional sprints delivering real-time sync and cross-platform compatibility.

Projects

- 1) **Multi-Agent RAG System with LangGraph** — *Agentic AI Orchestration* [Live Demo](#) Apr 2025 – Jun 2025
 - Architected a production-grade **multi-agent system using LangGraph** with a supervisor agent orchestrating specialized sub-agents: a retrieval agent (Pinecone hybrid search), a reasoning agent (GPT-4o), and a validation/grounding agent.
 - Implemented **tool-calling interfaces** with schema validation, retries, and fallback logic for safe and predictable agent behavior in production.
 - Integrated **LangSmith tracing** for full observability into agent reasoning chains; deployed behind FastAPI with Redis-based session memory enabling persistent multi-turn conversations.
 - Containerized with Docker Compose; integrated **GitHub Actions CI/CD** pipeline for automated testing and deployment gating.
- 2) **RAG-Powered Document Intelligence Chatbot** Mar 2025 – May 2025
 - Built a production-grade RAG chatbot using **LangChain and ChromaDB** for semantic Q&A over enterprise document repositories with GPT-4o as the reasoning backbone and custom prompt templates.

- Achieved <2s average response time on a 10K-document corpus; deployed as a **Dockerized FastAPI service** with Redis caching layer reducing redundant LLM API calls by **40%**.
- Implemented structured output parsing and conversation memory for a seamless, context-aware chat experience.

3) LangChain — Conversational AI & GenAI Application Development Ongoing 2025

- Built comprehensive LangChain expertise covering: LCEL chain architecture, prompt templates, structured output (Pydantic), output parsers (JSON, CSV), and custom formatters for reliable downstream integration.
- Implemented document loaders, text splitters, embedding pipelines, and vector store integrations (ChromaDB, FAISS, Pinecone) for production RAG systems over enterprise document corpora.
- Built functional conversational AI chatbots with persistent memory (**ConversationBufferWindowMemory**) and tool-calling agents via LangChain's AgentExecutor with custom tool definitions.

4) AI-Powered Lead Management System — *Full-Stack* Jan 2025 – Jun 2025

- Built a full-stack application using **React Native, FastAPI, and MongoDB** with JWT authentication, role-based access control, and real-time WebSocket sync.
- Designed RESTful APIs following OpenAPI spec; achieved **80% unit and integration test coverage**.

5) Scalable ML Inference Service — *Microservice Architecture* Jun 2024 – Aug 2024

- Designed a modular AI inference service using FastAPI, Redis (task queue), and Docker with hot-swappable model versions and A/B testing.
- Reduced average response time by **35%** via async batching on a Kubernetes cluster with Prometheus monitoring.

Skills

Conversational AI & LLM Engineering: LangChain (LCEL, Chains, Agents, Memory, Tools), LangGraph, Multi-Agent Orchestration, Tool Calling, Function Calling, MCP-style Interfaces, RAG Pipelines (Naive / Agentic / Hybrid), Prompt Engineering, Structured Output, LangSmith Observability

LLM APIs & Platforms: OpenAI API (GPT-4o, Embeddings, Function Calling), Anthropic API, Hugging Face Hub, Ollama, vLLM, LlamaIndex

Fine-Tuning & Evaluation: LoRA, QLoRA, PEFT, Instruction Tuning, GGUF/GPTQ, deepeval, ROUGE, BERTScore, LLM-as-Judge

Vector Databases & Embeddings: Pinecone, ChromaDB, Qdrant, FAISS, Sentence Transformers, OpenAI Embeddings, Hybrid Search (BM25 + Dense)

Software Engineering & APIs: Python, SQL, FastAPI, REST APIs, WebSockets, JSON, JWT, OpenAPI/Swagger, Git, CI/CD (GitHub Actions)

DevOps & Infrastructure: Docker, Kubernetes, Redis, MLflow, Weights & Biases, Prometheus, Streamlit, Bash

Full-Stack & Mobile: Android (Retrofit, REST API)

Databases: MongoDB

Soft Skills: Client-Facing Technical Delivery, Requirements Gathering, Solution Validation, Cross-Functional Collaboration, Agile/Scrum, Rapid Iteration

Education

B.Tech – CSE (IoT, Cyber Security & Blockchain Technology)	2022 – 2025
<i>Annasaheb Dange College of Engineering & Technology, Ashta</i>	<i>CGPI: 7.09</i>
Diploma in Computer Engineering	2019 – 2022
<i>Dr. Bapuji Salunkhe Institute of Engineering And Technology, Kolhapur</i>	<i>83%</i>